ANT-EON ZOO: THE NEURAL NEST

CAPABILTIES: THE FORWARD MARCH

LIMITATIONS: THE FIXED PATH

Structure	Mechanics	Applications
1. Unidirectional Flow Data moves strictly from input to output — no cycles or feedback loops. **2. Layered Architecture** Composed of an input layer, one or more hidden layers, and an output layer. **3. Fully Connected Neurons** Each neuron in one layer connects to every neuron in the next layer. **4. Activation Functions** Common choices include ReLU (Rectified Linear Unit), sigmoid, and tanh — used to introduce non-linearity. **5. Weight Parameters** Each connection has a weight that determines the strength of influence between neurons. **6. Bias Terms** Each neuron typically includes a bias to shift the activation threshold.	**1. Forward Propagation** Input data is passed through the network layer by layer, transformed by weights and activations. **2. Loss Function Calculation** The output is compared to the target using a loss function (e.g., mean squared error, cross-entropy). **3. Backpropagation** Gradients of the loss are computed with respect to each weight using the chain rule. **4. Gradient Descent Optimization** Weights are updated iteratively to minimize the loss — often using variants like SGD, Adam, or RMSprop. **5. Epoch-Based Training** The model is trained over multiple passes (epochs) through the dataset to improve accuracy.	**1. Binary Classification** Spam detection, fraud detection, medical diagnosis (e.g., disease presence). **2. Multi-Class Classification** Handwritten digit recognition, sentiment analysis, image labeling. **3. Regression Tasks** Predicting house prices, stock values, or energy consumption. **4. Feature Transformation** Used as a base layer in more complex models to extract meaningful representations. **5. Embedded Systems** Lightweight FNNs are often used in edge devices for fast, low-power inference.